Venkata Lakshmi Parimala Pasupuleti

AI Systems Engineer · Production LLM & Agentic Platforms · LLM Evaluation · Observability · Real Estate Domain · 9yr Production · Claude Code Daily

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA (Remote)

PROFESSIONAL SUMMARY

AI systems engineer who builds production LLM and agentic platforms. I built LeaseI: an 8-agent LangGraph pipeline taken from zero to production -- prompt design, orchestration architecture, evaluation harness, CI regression gate, iteration on real conversations. 93.6% precision across 1,240 compliance test cases. I built AgentProbe because I needed an evaluation framework to trust LeaseI before shipping prompt changes: 5-metric CI-integrated eval engine (correctness, groundedness, latency, token efficiency, hallucination rate) with LLM-as-judge scoring and a regression gate that blocks deploys on quality degradation. Claude Code is my primary daily development tool -- it's in First American's JD and it's how I build. I spent 8 years at Entrata owning compliance-critical systems for the real estate industry: eligibility, leasing, onboarding, recertification for 100K+ housing units. The domain First American's Title & Settlement Platform serves -- I've built production software in it. Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Agentic AI & LLM Systems: LangGraph (multi-agent orchestration), Agentic AI, Multi-Agent Systems, Production AI Systems, RAG Pipelines, Tool Orchestration, Function Calling, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Evaluation Framework Design, Observability, Agent Tracing, Confidence Scoring, HITL, Guardrails, Token Economics, Latency Optimization, Model Routing, Prompt Engineering, Context Management, LangChain, Embeddings, Document Intelligence, OCR, Claude Code (daily), Anthropic API, OpenAI API, Groq

Languages & Frameworks: Python (9yr production), TypeScript, JavaScript, SQL/PL-SQL, R, Bash; FastAPI, Flask, Angular, React.js, Django (familiarity)

Cloud, Backend & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, CI/CD (GitHub Actions), Cloud-Native Architecture, Scalable Backend Services, Fault-Tolerant Systems, PostgreSQL (200+ queries optimized, stored procedures, multi-tenant schemas), MySQL, Oracle, ETL/ELT Pipelines, RabbitMQ async processing, REST APIs, Microservices, Distributed Systems, Automated Testing, Regression Testing, Agile/Scrum, Git

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Real Estate Compliance & Operations Platform · Lehi, UT (Remote)

- **8 years owning compliance-critical systems in the real estate industry** -- same domain as First American's Title & Settlement Platform. Sole owner of Entrata's federal income-limit import and eligibility engine: annual ingestion of complex federal regulatory data expanded into ~180,000 domain records powering HUD/LIHTC/HCV eligibility, rent calculations, HAP payments, and recertification workflows for 100K+ housing units. Zero compliance violations. Pre-import validation, reconciliation, and rollback on every release because one wrong record means thousands of residents get the wrong eligibility decision.
- **Production engineering at scale** -- shipped 20+ full-stack production modules (Python, Angular/JavaScript, PostgreSQL) across 120+ Agile sprints; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual operations workload through Python/Bash automation pipelines; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and partitioning; RabbitMQ async processing with idempotency guarantees; AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents).
- **Engineering culture** -- mentored 5+ engineers; 300+ peer code reviews; built internal debugging tools reducing debugging time 40%; 99.5% uptime maintained across compliance-critical production systems.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and data engineering.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Systems & Regulatory RAG

- Built an agentic system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) over an FAA Part 107 regulatory corpus; RAG layer with cited-source compliance checks; validation guardrails enforcing regulatory constraints before export. End-to-end agentic workflow: planning loop, tool calls, failure mode handling, graceful degradation to rejection path on constraint violations.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · PostgreSQL · Real Estate Compliance

- Built property management and compliance modules; income-limit import for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific housing certification templates.

PROJECTS

Leasel: 0→1 Production Agentic Platform -- Full LLM Feature Lifecycle

LangGraph · Production AI · LLM Evaluation · Agentic Orchestration · HITL · FastAPI

Python · LangGraph (8-agent StateGraph) · FastAPI · TF-IDF RAG · SQLite · OCR · pdfplumber · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production LLM/agentic platform built 0→1 -- the full lifecycle First American's JD describes. **Architecture:** 8-agent LangGraph StateGraph with full response pipeline (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → HITL Escalation) -- planning loops, multi-step tool calls, conditional routing, graceful degradation (confidence drop → human escalation, never hallucinated compliance answer). **Retrieval:** TF-IDF RAG over complex regulatory corpus (deliberate choice over vector DB -- more reliable for compliance retrieval where exact clause matching matters; "pragmatic about models" = know when cheaper is also better). Section-aware chunking, missing/conflicting clause detection, 100% citation correctness. **Eval harness:** Built AgentProbe to instrument and gate this system before trusting it in production. **Results:** 93.6% precision, 87.9% F1, pass@1 81.7%, pass@3 88.3% across 1,240 test cases (5 US states + federal). CI regression gate blocks deploys on score degradation.

AgentProbe: CI-Integrated Evaluation Framework for Production AI Systems

LLM Evaluation · Hallucination Detection · Observability · CI/CD · Multi-Provider · React

Python · FastAPI · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic / OpenAI / Groq | github.com/venkatalakshmiparimala/AgentProbe

Evaluation framework and quality gate system for production AI -- directly what First American's "evaluation strategies, cost monitoring, latency budgets, and guardrails" requirement describes. **5-metric automated evaluation engine with quality metrics:** correctness, groundedness, latency, token efficiency, hallucination rate. **CI quality gates + regression gate:** GitHub Actions integration blocks deploys when quality scores degrade -- I don't ship a prompt change without knowing what it does across 1,240 conversations. **Observability:** structured agent tracing, confidence scoring, what was retrieved and what was ignored, where the agent branched -- gives CX teams an answer when a response goes sideways. **Operational AI:** multi-provider model routing (Anthropic, OpenAI, Groq) for cost management; latency and token efficiency tracked as first-class metrics. 15-endpoint FastAPI backend + React/TypeScript frontend.

AI on FHIR: Full Stack NLP-Driven Medical Query Platform

Flask · React · TypeScript · LLMs · HIPAA Compliance · Full Stack

Python · Flask · React · Docker · spaCy · BERT · Embeddings · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant Flask + React full-stack platform -- NLP + LLM + BERT translating natural-language queries (EN/ES) into FHIR API calls; RBAC across structured regulated records; reduced manual retrieval time ~60%. Demonstrates full-stack delivery and secure AI integration into regulated backend systems.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017